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Tree growth response to soil nutrients and neighborhood crowding varies between mycorrhizal types in an old-growth temperate forest

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Abstract

Forest dynamics are shaped by both abiotic and biotic factors. Trees associating with different types of mycorrhizal fungi differ in nutrient use and dominate in contrasting environments, but it remains unclear whether they exhibit differential growth responses to local abiotic and biotic gradients where they co-occur. We used 9-year tree census data in a 25-ha old-growth temperate forest in Northeast China to examine differences in tree growth response to soil nutrients and neighborhood crowding between tree species associating with arbuscular mycorrhizal (AM), ectomycorrhizal (EM), and dual-mycorrhizal (AEM) fungi. In addition, we tested the role of individual-level vs species-level leaf traits in capturing differences in tree growth response to soil nutrients and neighborhood crowding across mycorrhizal types. Across 25 species, soil nutrients decreased AM tree growth, while neighborhood crowding reduced both AM and EM tree growth, and neither soil nor neighbors impacted AEM tree growth. Across mycorrhizal types, individual-level traits were stronger predictors of tree growth than species-level traits. However, most traits poorly mediated tree growth response to soil nutrients and neighborhood crowding. Our findings indicate that mycorrhizal types strongly shape differences in tree growth response to local soil and crowding gradients, and suggest that including plant-mycorrhizae associations in future work offers great potential to improve our understanding of forest dynamics.

Keywords Functional traits · Mycorrhizal type · Neighborhood crowding · Soil nutrients · Tree growth

Introduction

Tree growth is a key component of forest dynamics (Clark and Clark 1999; Silvertown 2004), and a better understanding of its drivers can improve our ability to predict the future of forest ecosystems (Curzon et al. 2017). There is compelling evidence for the role of abiotic and biotic factors in shaping tree growth across tropical (Clark et al. 2003; Fortunel et al. 2018), subtropical (Liu et al. 2016), temperate (Canham et al. 2006) and boreal (Coates et al. 2013) forests. In particular, mycorrhizal fungi benefit their host plants by enhancing nutrient uptake (Smith and Read 2008) and stress resistance (Liu et al. 2018; DeForest and Snell 2020; Tedersoo et al. 2020), and a growing number of studies have evaluated their role in tree growth (van der Heijden et al. 2015; Jiang et al. 2020). However, most studies so far examined how plant species associated with different mycorrhizal partners (e.g. arbuscular mycorrhizal (AM) and ectomycorrhizal (EM) fungi) respond to either biotic or abiotic drivers (Lin et al. 2015; Bennett et al. 2017; DeForest and Snell

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2020), with few studies evaluating their response to abiotic and biotic factors simultaneously (Jiang et al. 2020). It thus remains unclear whether we can generalize how tree growth varies between mycorrhizal types along local abiotic and biotic gradients where different mycorrhizal types co-occur.

Soil nutrients represent a major abiotic constraint on tree growth in temperate forests (Miller 1981; Vitousek and Howarth 1991; Li et al. 2018) at both local (Zhao and Zeng 2019) and regional (Elser et al. 2007) scales. While host plants can generally obtain soil nutrients via the mycelial network of their mycorrhizal fungi, this benefit may differ across soil nutrient levels between mycorrhizal types (Tedersoo et al. 2020). Specifically, AM trees are favored in nutrient-rich soils because AM fungi can expand the volume of soil explored by short-lived hyphae, thus providing higher root foraging precision for host plants (Chen et al. 2018). In contrast, some EM fungi can mineralize soil organic matter by producing extracellular enzymes, making EM trees more competitive in nutrient-poor soils (Hodge and Fitter 2010; Phillips et al. 2013; DeForest and Snell 2020). In addition, AEM trees (i.e. trees hosting both AM and EM fungi) are generally found in similar environments as EM trees, and benefit from the complementarity in nutrient acquisition between AM and EM fungi, with limited cost to tree growth (Lambers et al. 2008; Plassard and Dell 2010; Teste et al. 2020). Most studies have focused on distribution differences between AM and EM trees along broad soil fertility gradients at global (Soudzilovskaia et al. 2015; Barceló et al. 2019; Steidinger et al. 2019) and regional (Bueno et al. 2017) scales, but we know relatively little about differences in tree growth between mycorrhizal types with local soil nutrient gradients.

Neighborhood interactions influence tree growth across tropical, temperate and boreal forests (Kunstler et al. 2016), via competition for shared resources (e.g. light) or attacks by shared natural enemies (Tilman 1982; Canham et al. 2006). Mycorrhizal associations can further mediate the response of tree growth to neighborhood interactions. In competitive environments, EM fungi constitute the extra-matrical mycelium that can increase soil nutrient acquisition for the host plant (Courty et al. 2010). In addition, EM fungi form a mantle around tree roots providing a physical barrier and can release antibiotic compounds slowing down pathogen sporulation, which can both act as protection against soil pathogens (Marx 1972; Duchesne et al. 1988; Courty et al. 2010; Bennett et al. 2017). As specific soil pathogens are more likely to accumulate around conspecific individuals (Chen et al. 2019), EM trees are therefore predicted to be less susceptible to neighborhood crowding effects than AM trees. The dual-mycorrhizal status of AEM trees may confer greater ability to exploit whole soil depth profile and better protection against pests and pathogens (Teste et al. 2017), which together may contribute to make them less vulnerable to neighborhood crowding. To date, most studies of the role of mycorrhizal associations have focused on seedling and sapling stages (Bennett et al. 2017; Jiang et al. 2020), with few empirical investigations in adult trees, especially in temperate forests.

Functional traits provide important insights into species strategies for resource use and biotic interactions (Westoby 1998; Westoby et al. 2002). They can capture species differences in tree growth and its response to soil and neighborhood crowding (Kunstler et al. 2016; Ruiz-Benito et al. 2017; Fortunel et al. 2018). For instance, species with acquisitive strategies (e.g. higher specific leaf area) have higher growth rates and are more sensitive to soil gradients and neighborhood crowding (Lasky et al. 2013; Fortunel et al. 2016). However, traits sometimes exhibit weak effects on tree growth (Poorter et al. 2008; Paine et al. 2015), which may in part be due to the use of species mean trait values in the analysis. Mean-trait approaches may indeed ignore important phenotypic plasticity between individuals, thereby limit the explanatory power of traits for tree growth (Long et al. 2011; Auger and Shipley 2013; Uriarte et al. 2016). Recent works in tropical and subtropical forests suggest that individual-level traits have a pivotal role in community dynamics (Liu et al. 2016; Umaña et al. 2018), but there is still limited evidence for the role of individual-level traits in shaping tree growth in temperate forests, in particular between tree species associated with different mycorrhizal partners.

Here, we used an unprecedented dataset of individual-level tree growth and leaf traits in an old growth, temperate forest in China, combined with detailed soil data and tree censuses, to test the relative influence of abiotic (soil nutrients) and biotic (neighborhood crowding) factors on tree growth. Specifically, we ask: (1) Do AM, EM and AEM trees exhibit growth differences across local soil nutrient gradients? As AM trees are more sensitive to the environment (Jo et al. 2019), we expect AM trees to grow faster than EM trees in nutrient-rich patches, with AEM trees showing an intermediate response. (2) Do AM, EM and AEM trees differ in their growth response to neighborhood crowding? Since EM fungi can increase soil nutrient acquisition and offer physical and chemical protection against soil pathogens, we expect EM trees to suffer less from neighbors than AM trees, with AEM responding even less to neighborhood crowding. (3) Are individual-level leaf traits better predictors of tree growth and its response to soil nutrients and neighbors than species-level leaf traits? We expect that individual-level leaf traits to better capture tree growth than species-level leaf traits.

Materials and methods

Study site

The study site was located in the Changbai Mountain Nature Reserve in Northeast China, extending from 41°43′ to 42°26′ N and 127°42′ to 128°17′ E. The reserve was established in 1960 and joined the World Biosphere Reserve Network in 1980. The area of this reserve is about 200,000 ha. The forest type is late-successional deciduous broadleaved Korean pine (*Pinus koraiensis*) mixed forest that has been undisturbed for at least 300 years, the most common vegetation type in the region. Common tree species include *Pinus koraiensis*, *Tilia amurensis*, *Quercus mongolica*, *Fraxinus mandshurica*, *Ulmus japonica*, and *Acer mono*. Mean annual precipitation is approximately 700 mm and most rainfall occurs from June to September (480–500 mm). Mean annual temperature is 2.8 °C, with a monthly mean of −13.7 and 19.6 °C in January and July, respectively (Wang et al. 2012).

Our study was conducted in the 25-ha Changbaishan (CBS) Forest Dynamics Plot (FDP), which is part of the Chinese Forest Biodiversity Monitoring Network and Forest Global Earth Observatory Network (ForestGeo). All individuals with a diameter at breast height (DBH) ≥ 1 cm in the CBS FDP were identified to species, tagged, measured and mapped. The forest was censused every five years since 2004. From the first census data (Appendix S1), there are 38,902 individuals belonging to 52 species, 32 genera and 18 families (Wang et al. 2015). In this study, we used the 2004, 2009, and 2014 census data.

Soil variables

In October 2007, surface soil samples (10 cm depth) were collected on a 30 m × 30 m grid in the CBS FDP. To capture variation in soil gradients at finer scales, two additional sample points were selected at 2, 5 or 15 m from each grid point in a random compass direction (Appendix S2). In total, 967 samples were analyzed for nine soil variables: total nitrogen (Ntot), available nitrogen (Nav), total phosphorus (Ptot), available phosphorus (Pav), total potassium (Ktot), available potassium (Kav), pH, organic matter (SOM), and water content (SWC) (see details in Yuan et al. (2011)).

We used ordinary kriging for 20 m × 20 m quadrats (Fig. 1a and Appendix S3) and performed a principal component analysis (PCA) of all nine soil variables to explore collinearity (Appendix S4 and S5). We found that the first two principal components, respectively, described 56% and 20% of observed soil variation at the 20-m scale. The soil PC_1 axis was strongly positively associated with Ntot, Nav, Pav, SWC and SOM, and negatively related with Ktot, while the soil PC_2 axis was strongly positively associated with Kav and Pav.

Individual tree growth

In August 2009, we installed dendrometer bands on 927 individual trees of 28 species, which belong to 17 genera and 12 families. Focal trees for each species were selected according to the first census data: all individuals with DBH ≥ 5 cm were divided into nine size classes (5–10 cm, 10–20 cm, 20–30 cm, 30–40 cm, 40–50 cm, 50–60 cm, 60–70 cm, 70–80 cm, and over 80 cm), and we randomly selected ten individuals from each size class. We used digital calipers (precision of ± 0.01 mm) measuring changes in dendrometer bands in May, September, and October of every year since September 2009. We calculated annual diameter growth $g_{i,t}$ (mm year^{-1}) for each tree i between each census year in the same month as:

$$g_{i,t} = \frac{\left(C_{i,t} - C_{i,t-1}\right)/\pi}{T_t - T_{t-1}},$$

where $C_{i,t}$ is the tree circumference in May at census year T_t, and $C_{i,t-1}$ is the tree circumference in May at census year T_{t-1}.

Here, we excluded (1) trees that died over the census interval; (2) trees whose dendrometer bands were damaged over the census interval; and (3) trees from species that have fewer than three individuals. In total, we analyzed nine years of tree growth for 709 individuals belonging to 25 species (Fig. 1b; Table 1).

Species mycorrhizal associations

We classified the 52 tree species found in the CBS plot into three mycorrhizal associations (AM/EM/AEM) following the literature (Wang and Qiu 2006; Guo et al. 2008; Brundrett 2009; Soudzilovskaia et al. 2020; and http://mycorrhizas.info/index.html), and identified 33 AM, nine EM and ten AEM tree species (Appendix S1). Amongst the focal 25 species with dendrometer data (Table 1), our dataset encompassed 12 AM tree species (317 individuals), seven EM tree species (250 individuals, among which 21.6% belong to two conifer species, and the remaining to five deciduous species), and six AEM tree species (142 individuals). Mycorrhizal types differed in tree size ranges (Table 1): AM trees had most small individuals (54%, DBH < 20 cm) and EM trees had most large individuals (58%, DBH > 40 cm).

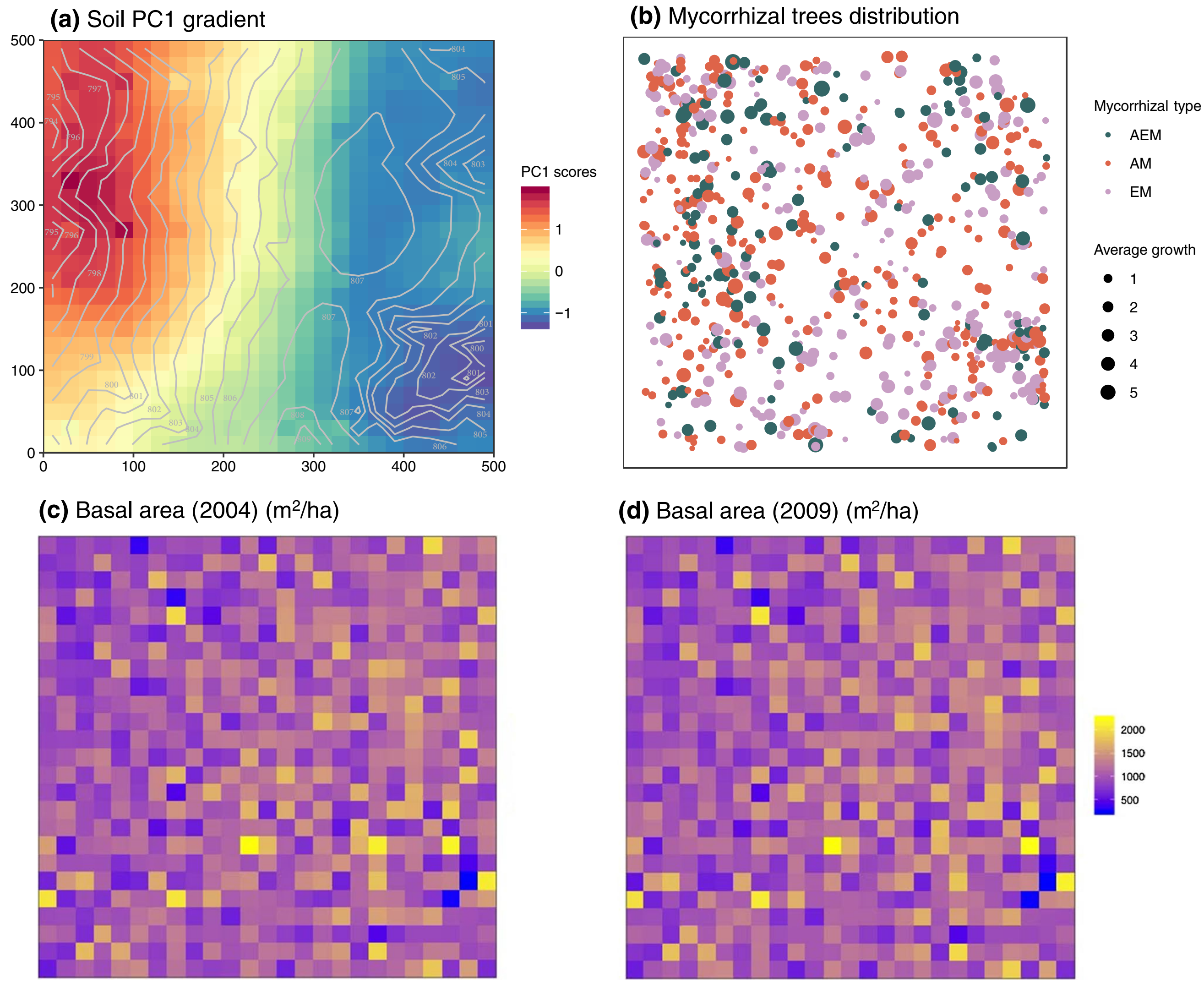

Fig. 1 Changbaishan Forest Dynamics Plot. **a** Soil PC_1 scores in each 20 m × 20 m quadrat (elevation lines are shown). **b** Mycorrhizal associations for individual trees with dendrometer bands (AM: arbuscular, AEM: dual, and EM: ectomycorrhizal). Point size indicates average tree growth during 2011–2019 period. Basal area in **c** 2004 and **d** 2009 in each 20 m x 20 m quadrat

Neighborhood crowding

In the CBS FDP, the total basal area in every 20 m × 20 m quadrat ranged from 214 to 2209 m^2 ha^{-1} during the first census, and from 237 to 2243 m^2 ha^{-1} during the second census (Fig. 1c, d). To evaluate neighborhood effects between two censuses, we calculated a neighborhood crowding index (NCI) for each focal tree at the beginning of each census interval, which was based on the size and distance of its neighbors within a 10 m radius:

$$\mathrm{NCI}_{s,i} = \sum_{k=1}^{K} \sum_{\substack{j=1 \\ j \neq i}}^{J_k} \frac{\mathrm{DBH}_{k,j}^{2}}{d_{k,j}},$$

where $d_{k,j}$ is the distance between focal tree i of species s and its neighbor j_k of species k, $\mathrm{DBH}_{k,j}$ is the size of the neighbor tree. Since the whole CBS FDP was censused at two dates overlapping the dendrometer surveys (2009 and 2014), we used the NCI_1 for tree growth data observed between 2009

Table 1 Characteristics of the 25 focal species at the Changbaishan plots that were monitored with dendrometer bands

Species	Family	Mycorrhizal type	Number of individuals	DBH mean [range] (cm)	Growth mean [range] (mm $year^{-1}$)
Populus koreana	Salicaceae	AEM	15	88.22 [40.2,142]	2.83 [0.90,4.97]
Populus davidiana	Salicaceae	AEM	9	19.46 [5.8,30]	1.36 [0.23,3.27]
Prunus padus	Rosaceae	AEM	21	14.2 [5.1,28.2]	1.39 [0.17,4.34]
Sorbus alnifolia	Rosaceae	AEM	12	10.51 [5.6,16.8]	1.09 [0.65,2.28]
Ulmus japonica	Ulmaceae	AEM	54	36.07 [5.4,86]	2.32 [0.15,5.91]
Ulmus laciniata	Ulmaceae	AEM	31	17.77 [5.8,53.2]	1.69 [0.30,4.22]
Acer tegmentosum	Aceraceae	AM	15	9.85 [5.7,14]	1.70 [0.70,3.34]
Acer mono	Aceraceae	AM	42	23.77 [5.2,46.4]	1.14 [0.09,3.98]
Acer mandshuricum	Aceraceae	AM	28	16.85 [5,33.8]	0.95 [0.07,2.74]
Acer triflorum	Aceraceae	AM	31	16.26 [5.4,33.4]	0.80 [0.12,2.37]
Acer pseudo-sieboldianum	Aceraceae	AM	24	12.6 [5,25.5]	0.71 [0.09,1.83]
Fraxinus mandschurica	Oleaceae	AM	55	40.24 [6.6,91.2]	2.56 [0.24,5.30]
Maackia amurensis	Leguminosae	AM	18	15.41 [6.3,27.9]	1.16 [0.37,2.01]
Malus baccata	Rosaceae	AM	48	18.34 [5.2,42.3]	0.80 [0.09,2.26]
Phellodendron amurense	Rutaceae	AM	21	22.45 [13.1,35.9]	2.17 [0.31,4.57]
Cerasus maximowiczii	Rosaceae	AM	9	9.42 [5.3,14.6]	0.64 [0.23,1.31]
Rhamnus ussuriensis	Rhamnaceae	AM	15	10.68 [6.2,18.5]	0.75 [0.19,1.34]
Syringa reticulata	Oleaceae	AM	11	9.47 [5.7,11.2]	0.62 [0.10,1.88]
Abies holophylla	Pinaceae	EM	6	43.55 [18.8,69.4]	3.50 [0.56,5.42]
Betula costata	Betulaceae	EM	8	31.48 [10.7,58.7]	2.58 [0.39,4.39]
Betula platyphylla	Betulaceae	EM	30	26.15 [9.5,49.9]	1.93 [0.11,4.05]
Pinus koraiensis	Pinaceae	EM	49	41.49 [8.4,98.1]	1.36 [0.06,4.09]
Quercus mongolica	Fagaceae	EM	64	42.5 [7.2,106.5]	1.93 [0.15,5.55]
Tilia amurensis	Tiliaceae	EM	49	43.58 [5.6,106.5]	1.46 [0.09,4.49]
Tilia mandshurica	Tiliaceae	EM	45	26.22 [5.5,52]	1.16 [0.11,2.66]
25	12	3	709	25.86 [5,142]	1.54 [0.06,5.91]

Species family, mycorrhizal type, number of monitored individuals, and their mean and range of diameter at breast height (DBH) and diameter growth are indicated

AM arbuscular mycorrhizal, *EM* ectomycorrhizal, *AEM* dual mycorrhizal

and 2013, and NCI_2 for tree growth data observed between 2014 and 2019.

Functional traits

We measured leaf traits for each of the 709 focal trees following standard protocols (Cornelissen et al. 2003; Gitelson et al. 2003; Pérez-Harguindeguy et al. 2013). We selected nine traits capturing important dimensions of plant strategies: leaf area (LA), specific leaf area (SLA), leaf dry matter content (LDMC), chlorophyll content (LChl), leaf carbon content (LC), leaf nitrogen content (LN), Foliar ^{13}C composition ($L^{13}C$), Foliar ^{15}N composition ($L^{15}N$), and leaf phosphorus content (LP) (Table 2).

For each focal tree, we selected three to five straight, healthy, and sun-exposed branches, and collected six to ten healthy and intact leaves for each branch. For LChl, we selected six leaves and used portable chlorophyll meters (CCM-300, Hudson, NH, USA) before averaging values. We selected ten leaves to weigh leaf fresh mass (JJ523BC, G&G, USA), and scanned the leaf area (Canon LiDE 110, Tokyo, Japan). Leaf dry mass was measured after drying at 60 °C for 48 h. SLA was calculated as the ratio of leaf area to dry mass. LDMC was calculated as the ratio of leaf dry mass to fresh mass. Oven-dried leaves were then ground to a fine powder (RETSCH MM400, GmbH, Haan, Germany). LC and LN were determined using an elemental analyzer (Vario EL III, Elementar, Hanau, Germany). $L^{13}C$ and $L^{15}N$ were analyzed using the isotopic Cavity-Ring Down Spectrometer with the Combustion Module (CMCRDS system, Picarro, CA, USA). LP was measured using PERSEE TU-1900 UV spectrophotometer (TU-1900, Beijing Persee Instruments Co. Ltd., Beijing, China) (see details in Fang et al. 2019). For each

Table 2 Leaf traits by mycorrhizal types at the Changbaishan plot

Trait	Abbreviation	Unit	Mycorrhizal type	Mean [Range]	Functions	References
Leaf area	LA	mm^2	AEM	22.58 [22.58,42.58]	Resource use	Givnish (1987) and Díaz et al. (2016)
			AM	18.87 [18.87,32.05]		
			EM	38.09 [38.09,53.42]		
Specific leaf area	SLA	$m^2\ g^{-1}$	AEM	44.87 [44.87,180.80]	Resource use and defense	Poorter and Bongers (2006)
			AM	28.09 [28.09,214.04]		
			EM	60.61 [60.61,157.50]		
Leaf chlorophyll content	LChl	$mg\ m^{-2}$	AEM	44.91 [44.91,432.91]	Resource use	Poorter and Bongers (2006)
			AM	49.36 [49.36,445.45]		
			EM	13.08 [13.08,442.50]		
Leaf dry matter content	LDMC	$g\ g^{-1}$	AEM	44.91 [44.91,432.91]	Resource use	Wilson et al. (1999)
			AM	49.36 [49.36,445.45]		
			EM	13.08 [13.08,442.50]		
Leaf carbon content	LC	$cg\ g^{-1}$	AEM	44.91 [44.91,432.91]	Resource use and defense	Wright et al. (2004)
			AM	49.36 [49.36,445.45]		
			EM	13.08 [13.08,442.50]		
Leaf nitrogen content	LN	$cg\ g^{-1}$	AEM	0.31 [0.31,2.21]	Resource use and defense	Wright et al. (2004)
			AM	0.43 [0.43,2.39]		
			EM	0.36 [0.36,2.29]		
Foliar ^{13}C composition	$L^{13}C$	‰	AEM	0.59 [0.59,-29.63]	Resource use	Farquhar et al. (1982)
			AM	1.02 [1.02,-29.74]		
			EM	0.26 [0.26,-29.61]		
Foliar ^{15}N composition	$L^{15}N$	‰	AEM	0.56 [0.56,-0.90]	Resource use and symbionts associations	Bonan (2008)
			AM	1.13 [1.13,-1.18]		
			EM	0.32 [0.32,-0.73]		
Leaf phosphorus content	LP	$\mu g\ g^{-1}$	AEM	0.29 [0.29,1.79]	Resource use	Reich and Oleksyn (2004)
			AM	0.32 [0.32,1.79]		
			EM	0.26 [0.26,1.81]		

Trait mean and range (minimum and maximum) are indicated for each mycorrhizal type

AM arbuscular mycorrhizal, *EM* ectomycorrhizal, *AEM* dual mycorrhizal

species, we calculated mean trait values across all sampled individuals.

We ran a PCA for all nine functional traits at the individual level and species level, and examined differences between the three mycorrhizal types (Appendix S6, S7 and S8). The first two PC axes together explained ca. 40 and 47% of the variation at the individual level and species level, respectively. The trait PC_1 axis contrasted trees with high SLA and LN and low LDMC, while the trait PC_2 axis was associated with $L^{13}C$ and LChl.

Analyses

We build a Bayesian model of individual tree growth as a function of trait, tree size (DBH), soil nutrient, neighborhood crowding (NCI) and their interaction with traits as follows:

$$\begin{aligned}\log(g_{t,s,i}) = {} & \beta_{1s} + \beta_{2s}\,\mathrm{trait}_{s,i} + \beta_{3s}\log(\mathrm{DBH}_{t,s,i}) \\ & + \beta_{4s}\,\mathrm{soil}_{s,i} + \beta_{5s}\log(\mathrm{NCI}_{c,s,i}) \\ & + \beta_{6s}\log(\mathrm{DBH}_{t,s,i})\,\mathrm{trait}_{s,i} + \beta_{7s}\,\mathrm{soil}_{s,i}\,\mathrm{trait}_{s,i} \\ & + \beta_{8s}\log(\mathrm{NCI}_{c,s,i})\,\mathrm{trait}_{s,i} + \delta_t + \gamma_i + \eta_m\end{aligned}$$

where for a given individual i of species s at census t, $g_{t,s,i}$ is the growth rate between census t and $t-1$, trait $_{s,i}$ is the trait value individual i of species s, $\mathrm{DBH}_{t,s,i}$ is the diameter at 1.3 m height at census t, $\mathrm{soil}_{s,i}$ is the soil variable of the 20 m × 20 m quadrats where the focal tree is located, and $\mathrm{NCI}_{c,s,i}$ is the neighborhood crowding index in a 10 m radius at the start of the census interval (either 2009 or 2014). The parameter β_{1s} represents the intercept (i.e. the average tree growth of species s), the parameters β_{2s}, β_{3s}, β_{4s} and β_{5s} characterize the growth response of species s to trait, tree size, soil and neighbors, and the parameters β_{6s}, β_{7s} and β_{8s} characterize the growth response of species s to the interactions between trait with tree size, soil and neighbors. Fixed effect for the year is denoted as δ_t, and random effects for individual trees and quadrats are denoted as γ_i and η_m. For each covariate $c \in (1, 2, \ldots, 8)$, the species-specific parameters β_{cs} were modeled as arising from a normal distribution with mean β_0 and standard deviation $\beta_{c,0}$.

We standardized all model covariates to enable direct comparisons of their relative importance. We standardized soil variables by subtracting the mean and dividing by one standard deviation computed across all plot quadrats. DBH and NCI were both log-transformed prior to standardization. We standardized DBH and NCI by subtracting species-specific means and dividing by species-specific standard deviations. All functional traits were log-transformed to prevent the tail of their skewed distribution from dominating the model fit. $L^{13}C$ and $L^{15}N$ have negative values, so we added a minimum value before logging. Then we scaled all traits with mean = 0 and standard deviation = 1.

To manage model complexity and correlations among mycorrhizal association and leaf traits, we fit the growth models separately for each of the four tree species category (all 25 species, AEM species, AM species and EM species), thus allowing each mycorrhizal type to behave differently. We used all possible combinations of (1) each of the six soil variables (quadrat values of soil Nav, Pav, Ntot and Ptot, and quadrat scores on soil PC_1 and PC_2), and (2) each of the nine leaf traits (LA, SLA, LChl, LDMC, LC, LN, $L^{13}C$, $L^{15}N$ and LP). In addition, to compare the contribution of individual-level and species-level traits, we built separate models using either (i) species means or (ii) individual traits. Finally, to test whether a larger radius would better capture neighborhood effects on tree growth, we calculated NCI at a 20 m radius, and fitted tree growth model, using soil PC_1 and individual-level traits. We found similar results using NCI at a 10 or 20 m radius (Appendix S9), so we hereafter present models using NCI at a 10 m radius to allow direct comparisons with similar forest studies.

All parameters were set proper, diffuse priors and posterior sampling using Markov Chain Monte Carlo sampling techniques in JAGS 3.4.0 (Appendix S10). For each model, we ran 80,000 iterations with a burn-in period of 20,000 iterations, and the Gelman and Rubin diagnostics with a threshold value < 1.01 to assess parameter convergence. We computed the median and 95% credible intervals of model parameters (Appendix S11) and evaluated model goodness of fit (Appendix S12). All analyses were conducted in the R 3.6.2 statistical platform using packages r2jags (Su and Yajima 2015) and smart (Warton et al. 2012).

Results

Differences in tree growth between mycorrhizal types

Across the 709 focal individuals, average tree growth was 1.54 mm $year^{-1}$. The six AEM trees had a higher average growth rate in our data (1.94 mm $year^{-1}$), following by the seven EM trees with an average growth rate of 1.65 mm $year^{-1}$. The 12 AM trees were the slowest growing tree species with 1.30 mm $year^{-1}$ (Table 1).

Influences of tree size, soil and neighbors on tree growth between mycorrhizal types

Overall, tree growth increased with greater tree size across all focal species ($\beta_{3,0}$ parameter, Fig. 2a, Appendix S13). Considering the mycorrhizal association, tree growth of AM and AEM trees increased with tree size, while EM trees showed no significant growth response to tree size (Fig. 2b–d, Appendix S13). Using fitted model parameters to compare tree growth at similar tree size between mycorrhizal types, we found consistent response of tree growth to DBH across the range of tree size observed in our plot, with no reversal between AM, AEM and EM trees (Appendix S14).

Contrary to our expectation, considering all species together, tree growth overall had no consistent and relatively weak response to our studied soil nutrients ($\beta_{4,0}$ parameter, Fig. 2a, Appendix S11 and S13). Specifically, tree growth tended to be slower with increasing soil PC_1, Ntot and Ptot. Mycorrhizal types showed strong differences in tree growth response to soil: AM trees grew less with increasing soil nutrients (e.g. PC_1, Nav, Ntot and Ptot), while EM and AEM trees showed no growth response to soil nutrients (Fig. 2b–d, Appendix S11 and S13).

Tree growth decreased with increasing crowding by neighbors ($\beta_{5,0}$ parameter, Fig. 2a). As predicted, this effect was particularly marked in AM trees, while it was weak in EM and not significant in AEM trees (Fig. 2b–d, Appendix S11).

Role of individual-level versus species-level leaf traits between mycorrhizal types

Species with different mycorrhizal types differed in their leaf traits (Table 2, Appendix S6). Considering individual-level traits, AM trees exhibited higher SLA and lower LDMC (trait

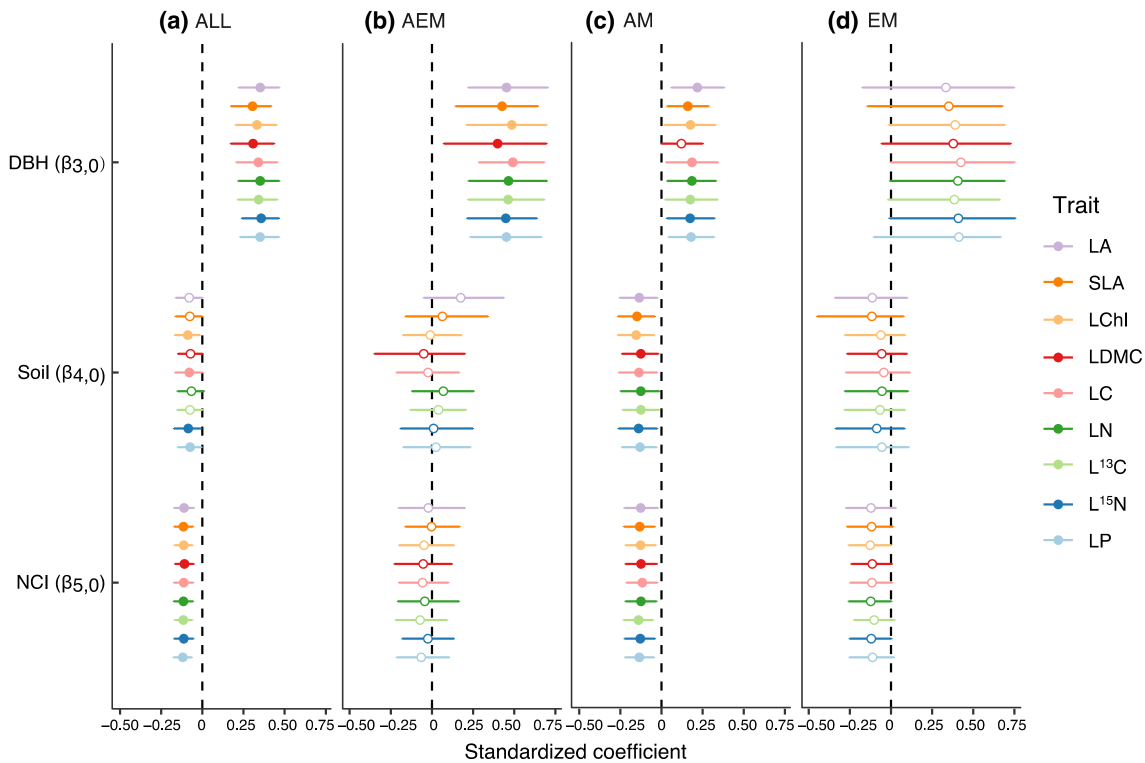

Fig. 2 Standardized regression coefficients modeling the effects of tree size ($\beta_{3,0}$), soil nutrient ($\beta_{4,0}$), neighborhood crowding ($\beta_{5,0}$) on tree growth, using tree DBH, soil PC_1 and NCI at 10 m radius, for **a** all 25 focal tree species, **b** AEM trees, **c** AM trees and **d** EM trees. Model parameter median values and 95% confidence intervals are shown for each species-level trait. See Table 2 for trait abbreviations

PC_1), and lower $L^{13}C$ and LChl (trait PC_2) than EM trees, while AEM trees occupied an intermediate trait space. For species-level traits, the contrasts between AM and EM trees were even stronger, but EM and AEM shared the same trait space. In sum, EM and AEM trees appeared to have relatively conservative strategies for light capture and nutrient use (trait PC_1), and higher photosynthesis and water use efficiency (trait PC_2) than AM trees.

In line with our prediction, individual-level traits were stronger predictors of tree growth than species-level traits (Fig. 3; Appendix S15). Across all tree species, species with lower SLA, and greater LA and LChl had faster tree growth ($\beta_{2,0}$ parameter, Fig. 3a). Importantly, traits predicted tree growth only in AM trees (Fig. 3b–d). Specifically, AM trees with greater individual-level LA, LChl, and LDMC grew faster ($\beta_{2,0}$ parameter, Fig. 3c).

Contrary to our prediction, traits did not consistently mediate tree growth response to tree size, soil nutrients and neighbors (Appendix S16). We found some limited evidence that tree growth further decreased with tree size in species with greater SLA, LChl, and $L^{13}C$ ($\beta_{6,0}$ parameter, Appendix S16a), while tree growth increased with increasing soil PC_1, Ntot and Ptot on AM species with greater LDMC ($\beta_{7,0}$ parameter, Appendix S16c).

Discussion

Examining differences in tree growth between mycorrhizal types with tree size, soil nutrients and neighborhood crowding can shed light on the relative contribution of different mechanisms driving forest dynamics. At the CBS plot, located in an old-growth temperate forest in Northeast China, we found that (1) tree growth increased with tree size, in particular in AM and AEM trees, (2) tree growth did not vary with local soil nutrient gradients, though AM trees grew slower with increasing soil nutrients, and (3) tree growth decreased with increasing neighborhood crowding, with a stronger impact on AM trees than EM and AEM trees. Overall, leaf traits did not mediate tree growth response to tree size, soil nutrients or neighbors, but individual-level traits better captured differences in tree growth than species-level traits, especially in AM trees.

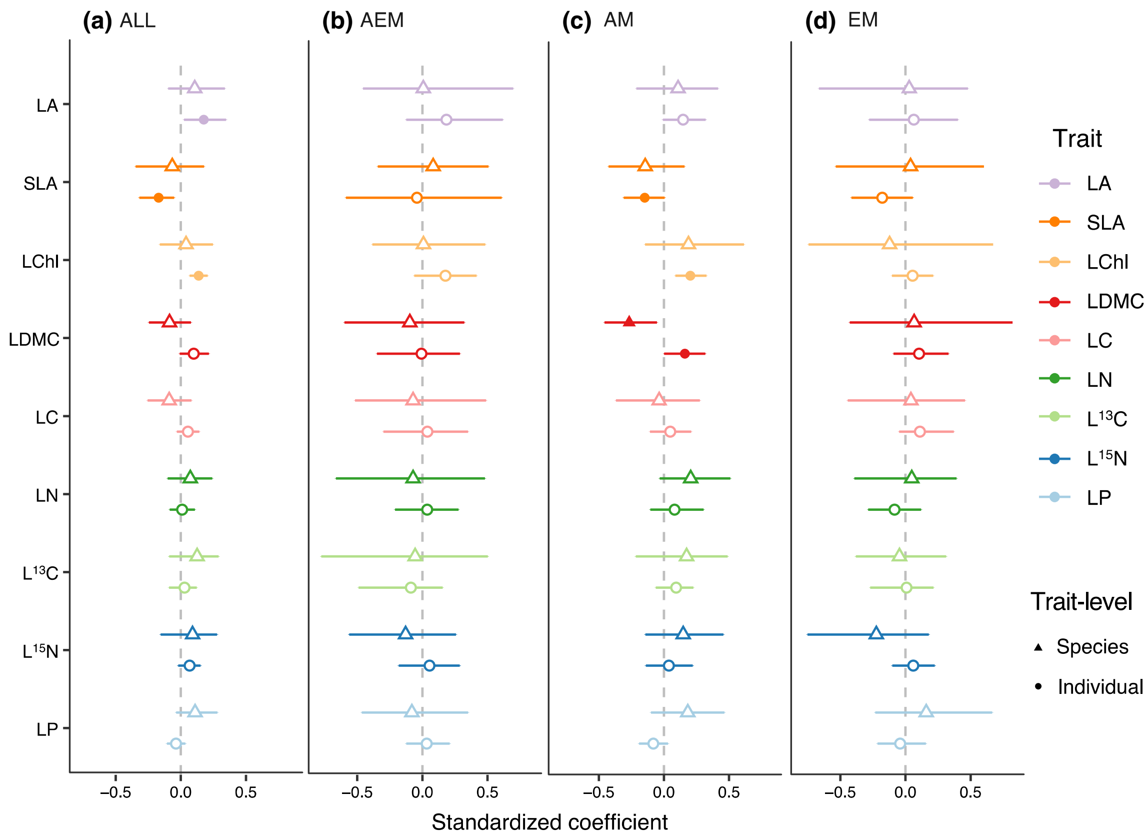

Fig. 3 Standardized regression coefficients of traits effects ($\beta_{2,0}$) on tree growth for species-level (triangle) and individual-level (circle) traits, using soil PC_1 and NCI at 10 m radius, for **a** all 25 focal tree species, **b** AEM trees, **c** AM trees and **d** EM trees. Model parameter median values and 95% confidence intervals are shown for each trait models. See Table 2 for trait abbreviations

Soil nutrients effects

In temperate forests, there is both direct (e.g. nitrogen addition experiments) and indirect (e.g. studies of element ratios) evidence of tree growth limitation by soil nitrogen and phosphorus (Miller 1981; Li et al. 2018). Here, we, however, found no community-level tree growth response to soil available nitrogen and phosphorus, and inconsistent response to soil total nitrogen and phosphorus. These limited effects may stem from the overrepresentation of large stems (> 10 cm DBH) at the CBS plot (79.8% of studied trees), as bigger trees may be less responsive to soil nutrients (Li et al. 2018). In addition, tree growth may be limited by other soil nutrients such as potassium that relate to leaf chlorophyll content and photosynthetic rate (Erel et al. 2015). In our study, total potassium was negatively linked with the first soil principal component, and available potassium was positively associated with the second soil principal component. Yet, we found no clear effects of either on tree growth. Future studies would need to include other potential limiting elements to clarify how abiotic factors influence tree growth.

Mycorrhizal fungi generally increase soil nutrients uptake and subsequent growth of their host plants (Phillips et al. 2013; Liu et al. 2018). Fungal hyphae extend the soil volume that plants can explore for soil nutrients, and EM fungi produce enzymes that allow plant access to soil organic N (Lindahl and Tunlid 2015). While previous work used broad soil fertility gradients to analyze growth variation of host plants associated with a particular mycorrhizal type (Püschel et al. 2016; Ma et al. 2021), we here used local soil gradients in a forest plot where all mycorrhizal types co-occur. Contrary to our prediction, we found that AM trees grew slower with increasing soil nutrients, while AEM and EM trees showed no response. In particular, we detected no effect of soil available and total P on AM tree growth, which contradicts studies conducted at single sites (Liu et al. 2018) but is consistent with a recent global meta-analysis (Ma et al. 2021). Our results may be linked to the negative effect of nitrogen and phosphorus fertilization on AM fungi abundance (Treseder 2004), which may slow down AM tree growth. Moreover, lower soil pH with long-term fertilization can limit AM hyphal development and root colonization, which may in turn decrease AM tree growth (Coughlan et al. 2000). An interesting future step would evaluate the relative influence of soil macronutrients and micronutrients on tree growth between mycorrhizal types.

Neighborhood crowding effects

Neighborhood crowding reduces tree growth across different forest biomes (Canham et al. 2006; Uriarte et al. 2010; Fortunel et al. 2018). Similarly, we found that neighbors reduced community-level tree growth at the CBS plot. In addition, we showed steeper growth reduction in AM trees than EM trees, in line with other studies in temperate forests (Bennett et al. 2017; Brown et al. 2020). AM fungi can provide some protection against soil pathogen attacks to their host plants (Liang et al. 2015), but EM fungi may provide greater physical and chemical protection for their host plants (Marx 1972; Courty et al. 2010; Bennett et al. 2017; Brown et al. 2020). Our results overall suggest that EM trees are less sensitive to competition for resources and shared natural enemies than AM trees (Laliberté et al. 2015). Furthermore, we found that AEM trees showed no response to crowding, in line with the expectation that having both mycorrhizal types confers improved soil nutrient access and protection against soil pathogens (Teste et al. 2017). Our study on adult trees complements previous studies that mainly focused on seedling and sapling stages, and provides further evidence that neighborhood crowding effects on tree growth varies between mycorrhizal types. An interesting next step as data on soil pests and pathogens becomes available would be to explore the relative role of con- and hetero-mycorrhizal neighbors on tree growth, in particular to shed light on negative-density dependence aspects. Our study focused on tree growth, but a better understanding of the role of mycorrhizal fungi in shaping forest dynamics and coexistence would need to combine multiple vital rates (including mortality, reproduction and recruitment).

Role of individual- versus species-level leaf traits

Most neighborhood crowding studies so far used species-level traits (Uriarte et al. 2010; Lasky et al. 2013; Fortunel et al. 2016), but there is a growing interest in evaluating the role of individual-level traits (Yang et al. 2021). At the CBS plot, we found that individual-level traits were stronger predictors of tree growth than species-level traits, consistent with recent findings in subtropical and tropical forests (Liu et al. 2016; Yang et al. 2021). In particular, trees with higher photosynthesis rate (LChl) and light capture surface (LA) grow faster, as they are better at capturing light (Givnish 1987; Poorter and Bongers 2006). Though SLA is usually associated with light use efficiency (Wright et al. 2004), we found that species with high SLA grew less, which may stem from contrasted growth and traits patterns between angiosperms and conifers at our site: our fastest growing species is a conifer (*Abies holophylla*) with low SLA, while slow-growing species are mostly deciduous, understory species with high SLA. With mounting evidence for the role of intraspecific variation in tree response to climate change (Anderegg and HilleRisLambers 2016; Johnson et al. 2018; McGregor et al. 2021), our study highlights that a promising way towards a better mechanistic understanding and thus forecasting of individual performance with global change drivers is to leverage individual-level trait information.

Interestingly, leaf traits significantly influenced tree growth only in AM trees at the CBS plot. Evaluating differences in leaf trait space between mycorrhizal types, AM trees showed wider ranges of trait values and exhibited higher values of LN (trait PCA_1) and LChl (trait PCA_2) than EM trees. This result stands in line with recent work showing that AM trees have acquisitive traits, while EM trees exhibit conservative traits (Averill et al. 2019). Our findings thus further suggest a stronger relationship between leaf traits and tree growth in trees with acquisitive strategies (Poorter et al. 2008).

Species trait differences can explain tree growth variation along abiotic and biotic gradients (Lasky et al. 2014; Fortunel et al. 2018). In contrast, we found that our focal nine leaf traits hardly mediate species growth response to soil nutrients and neighbors at the CBS plot. We showed that only AM trees with higher individual-level LDMC grew faster with increasing soil nutrients, which indicates the potential for leaf structural traits to reflect nutrient acquisition and growth investment (Yang et al. 2018). To improve our understanding of the role of soil nutrients and pathogens in forest dynamics, a promising way forward would be to integrate root traits (Chen et al. 2016), e.g. specific root length as it relates to greater soil phosphorus acquisition and lower defense against soil pathogens (Eissenstat 1992), and root nitrogen at it can capture tree growth response to soil nutrients (Laliberté et al. 2015; Chen et al. 2018).

Conclusions

Using individual-level dendrometer growth data for different mycorrhizal trees (AM, EM and AEM) in an old-growth, temperate forest in Northeast China, we find that soil nutrients had a limited, negative effect on tree growth, in particular in AM trees, while neighborhood crowding consistently reduced tree growth, with a more pronounced effect in AM trees. In addition, we find individual-level traits are stronger predictors of tree growth than species-level traits, especially in AM trees. Our study highlights the importance of integrating information on tree mycorrhizal associations and individual-level traits to improve our understanding of the drivers of tree growth and forest dynamics.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00442-021-05034-2.

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Author contribution statement JR, ZQH and CF designed the study. JR, SF, GL, FL, ZQY, JY, XGW and ZQH collected the data. JR, SF and CF performed the analysis. JR and CF wrote the first draft of the manuscript. All authors contributed to revisions of the manuscript, and gave final approval for publication.

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Data availability Data are available upon reasonable request to ZQH.

Declarations

Conflict of interest All authors declare no competing financial interests.

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